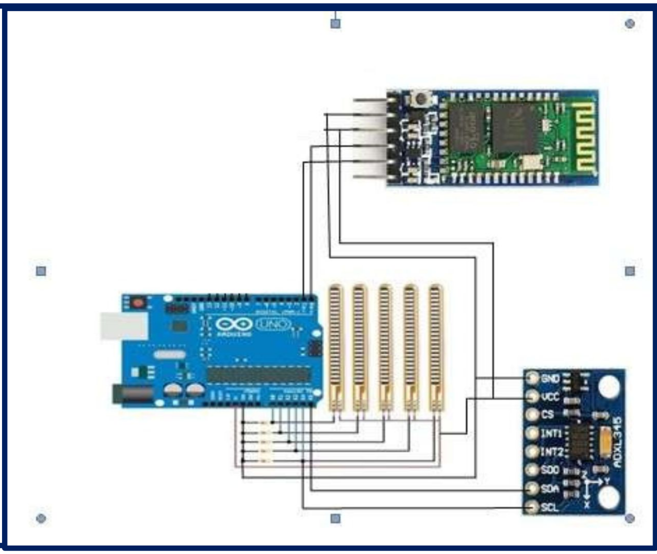
UE23CS251B: MPCA 4th Semester Section

Smart Glove-Based Sign Language Detection System

In this project, we present a smart glove system capable of recognizing sign language gestures using four flex sensors and an MPU6050 motion sensor. The glove captures finger bending (via flex sensors) and hand orientation/movement (via the MPU6050), and transmits the data wirelessly using an HC-05 Bluetooth module. These sensor readings are processed to detect specific hand gestures, which are then converted into corresponding text outputs. The system provides a low-cost, real-time solution to bridge communication gaps between speech-impaired and hearing individuals, emphasizing portability, accuracy, and ease of use.



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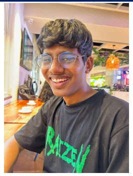

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